PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: KreditX

1. Project Overview

- **Problem Statement:**
  Being an entrepreneur requires necessary skills and knowledge in order to build a business which is able to generate profit. Many individuals who are in rural areas running small micro businesses such as Handicraft, Food Stalls, Home based Bakeries especially womens, often find it difficult to get loans. This is mostly due to not having a credit history as all the information is either paper-based or using whatsapp Chats which is not a solid evidence to get a loan.

- **Target Domain:** Unbanked rural individuals especially woman who need to take a loan to either upgrade their stall or to buy necessary equipment, inventory, etc.

- **Proposed Solution Summary:** KreditX is a User friendly platform that even allows an individual with no prior knowledge to use it easily. This decision Intelligence is able to explain decisions clearly with proper guidance and it is able to handle and interpret mixed data structures including the simple structures as well.

2. Background & Business Objective

- **Background of the Problem:** In rural areas, many unbanked entrepreneurs run small micro-businesses such as food stalls, home-based bakeries, or handicraft enterprises. Based on existing studies on entrepreneurial challenges and financial access, they often struggle with microloan applications because they do not have a formal credit history. They use informal records such as WhatsApp chats, voice notes, or paper notes to communicate with customers and record their orders. Consequently, loan officers rely on guesswork rather than data, leading to rejection of these applications. As a result, these

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people stay stuck in low-income cycles, unable to invest in inventory, equipment, or marketing to grow their businesses.

- **Importance of Solving This Issue:** By turning informal financial behaviours into meaningful credit insights, loan officers can make decisions based on structured data, which is more reliable than guesswork-based judgment. This improves the fairness of microloan approvals and reduces rejections caused by informal business records. It also empowers unbanked entrepreneurs to make better financial decisions, break out of low-income cycles, and gain opportunities to invest in their businesses.

- **Strategic Fit / Impact:** It aligns with the growing trend of AI-powered decision intelligence systems. The system helps in areas such as risk scoring, pricing and structuring, growth strategy recommendations, and explainable outputs that translate dialects into plain language. Gemini 2.5 Flash model plays a vital role to interpret unstructured data such as WhatsApp chats and voice notes, and simple data such as spreadsheet-style daily income and expenses. It integrates these inputs into a unified decision-support system that enables data-driven financial decision-making. Without the Gemini Flash model, the system would lose its ability to generate meaningful insights from unstructured data.

3. Product Purpose

3.1. Main Goal of the System

To provide an AI-powered decision intelligence partner for unbanked rural women entrepreneurs. This system could help them organise informal records such as WhatsApp chats, voice notes, or paper notes into clear creditworthiness insights and provide fair microloan recommendations. Although without formal financial knowledge or higher education, users can rely on the system to access financing and grow their micro-business with confidence.

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3.2. Intended Users (Target Audience):

Primary users: End-users

- Unbanked entrepreneurs and other individuals in rural Malaysia, who run micro-businesses such as handicraft sellers, food stall owners, or home-based bakeries.
- Low-income (B40) individuals with limited formal education and no prior experience with banking or digital finance tools.

Secondary users: Stakeholders or Decision-makers

- Community lenders and microfinance officers who assess loan applications for rural entrepreneurs.
- Field agents or cooperatives that support rural entrepreneurs with financial services.
- SME or microfinance program officers in government agencies and NGOs working with women-led businesses in rural areas in Malaysia.

4. System Functionalities

4.1. Description: The system operates as an AI powered decision intelligence engine for credit analysis. It bridges the gap between informal business practices such as WhatsApp chats, voice notes, paper records and formal credit assessment. Using a combination of natural language understanding and structured data processing, the system transforms scattered, unstructured financial traces into clear, explainable creditworthiness insights. It then provides microloan recommendations and growth guidance in plain, simple language to the user.

4.2. Key Functionalities:

- **Vibe Designing:** The system can translate unstructured data into structured data that can be used for credit analysis. It can accept voice notes, WhatsApp conversations with customers, photos of handwritten sales logs and text

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descriptions. Then the AI interprets and extracts key business data such as daily revenue, order frequency and seasonal demand.

- **Conversational Editing:** Users with low digital literacy can speak or type in their local dialect such as Malay and English to provide necessary details .
- **Versioning & Infinite Canvas:** The system generates a risk score, qualification status, loan recommendation and credit improvement strategy based on structured and unstructured data.
- **Export- Credit Analysis Report:** With one click, users are able to download the assessment report in pdf format. The report includes business activity summary, estimated daily revenue, financial summary, calculated creditworthiness score and supporting evidence snippets. The file can then be shared digitally with microfinance officers.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: We have used the Gemini well known LLM model which is known as Gemini 2.5 Flash .This model was chosen based on its capability where it is able to handle high-volume enterprise tasks with high performance and efficiency , reducing the overall waiting time of the user. As our system gets image, voice and audio input it is essential that those elements are processed well in order to send the right information therefore this specific model is able to handle this with very minimal complexity .

4.3.2. Prompting Strategy: Our system uses a few-shot prompting strategy combined with structured instructions prompting. In this approach the model is provided with clearly defined input fields (e.g. business type, years of operating), and has a small number of example scenarios with both approved and rejected cases. Clear and explicit instructions in order to provide a structured decision for the output.

This strategy is suitable for our case as loan evaluation requires consistent and explainable reasoning , rather than creative generations. The few-shot examples helps the model to always

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learn about the decision patterns involved such as being able to identify how low profit means higher rejection risk, short business duration shows instability and high loan to income ratio presents the repayment risks. By this method it significantly reduces hallucination and improves decision reliability compared to zero-shot prompting.

4.3.3. Context & Input Handling: This system collects structured inputs via a streamlit form, ensuring that most of the data is already organized. As the inputs are form-based they are naturally limited, estimated safe input range around 1000-2000 characters in total. The system would give a user friendly message if the user did not fill in the information or they have filled in with wrong type of data.

4.3.4. Fallback & Failure Behavior: the system includes safeguards for handling invalid off-topic or the so-called hallucinated responses from the model. The mechanisms that have been implemented are output validation , the response is checked for relevance to loan evaluation and if its not it is flagged automatically. Retry mechanism where the system is able to retry if any sort of error occurred during processing.

5. User Stories & Use Cases

- **User Stories:**
    1) As a small business owner, I want to enter my business and financial details so that I can know if I am eligible for a loan as its very hard to understand what needs to be prepared or done to have a loan applied .
    2) As a user, I want a simple interface to input my data so that I can complete the loan evaluation quickly without having to understand a sophisticated user interface that leads to more confusion.
    3) As a user, I want to receive a clear loan decision (approved/rejected) so that I can understand my financial standing without having any prior knowledge about finance .

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- **Use Cases (Main Interactions):**

  1)**User Input to UI flow:** Users access the application through the Streamlit interface and input required business and financial details (e.g., business name, income, expenses, loan amount).

  Once submitted, the frontend sends the data to the FastAPI backend endpoint (/evaluate_loan) for processing. The backend validates the input and forwards it to the AI/LLM model for evaluation.

  2) **Loan Evaluation Flow :** After receiving the input, the backend processes the data using the loan evaluation logic (LLM-based assessment). The system analyzes financial stability, repayment ability, and risk factors. The AI model then generates a structured response including, loan approval status, risk level and explanation/reasoning

  3. Response Rendering Flow: The backend returns the evaluation result to the Streamlit frontend. The UI then dynamically displays the result in a user-friendly format, showing approval/Rejection status, key insights or reasons and finally any additional recommendations (if applicable)

- **Use Case Diagram:** *(Optional but encouraged for participants to visualize the interaction)*.

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6. Features Included (Scope Definition)

The purpose of this system is to address rural area entrepreneurs' challenges in securing loans by converting informal records into credible credit profiles through AI-powered decision intelligence. The features focus on simplicity and usability for low-tech users.

- **Core Interface:** A conversational WhatsApp powered by Gemini LLM (gemini-2.5-flash) supports natural chats in Bahasa Malaysia or mixed English. Users can easily submit voice notes, photos of paper records, and text inputs without needing apps or literacy skills.
- **Data Processing:** AI-driven organises mixed data from WhatsApp chats, voice notes, and scanned paper notes into structured credit profiles. This includes clear risk insights and assessments of business stability.
- **Decision Outputs:** The system generates explainable credit scores, recommended microloan amounts, and stakeholder suggestions in simple language, with visualisations like charts. It also generates explainable outputs for stakeholders, highlighting user credit profiles and key risk factors.
- **Stakeholder Access:** A secure stakeholder dashboard with a simple login function and exportable PDF reports of user profiles. This enables quick reviews for microfinance institutions.
- **Design Standards:** The coherence of the UI interface in both Bahasa Malaysia and English is consistent across user and stakeholder flows. It also supports export of credit reports, including shareable codes or links for banks and external stakeholders.

7. Features Not Included (Scope Control)

- Real WhatsApp Business API integration (demo uses simulated chat interface)
- Full user authentication and role-based access control

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- Live integration with real microfinance institutions or banks

8. Assumptions & Constraints

- **LLM Cost Constraint:** We estimate an average of 3,500–4,500 tokens per user session. We assume a typical interaction including uploading 3 to 5 WhatsApp screenshots, photos of handwritten records, voice notes, a brief conversational Q&A and generating a loan recommendation with explanation. The estimated LLM cost is around RM 0.03 per session.

  We implement prompt caching for repeated business patterns — for example, when a user submits similar daily sales entries (e.g., "sold 10 packs today like yesterday"), the system reuses previous inference results instead of reprocessing from scratch. Additionally, we limit unstructured data processing to the most recent 30 days of chat history per session, since older records provide diminishing returns for credit assessment.

- **Technical Constraints:** The system generates the credit analysis report in PDF format only.

- **Performance Constraints:** Each session can process a maximum of 1 image. Voice note processing has a 60-second limit per upload.

- **User Input:** All loan recommendations are decision-support only. Final approval requires review by a microfinance officer or community lender as the system is not a licensed credit authority. Users must confirm the accuracy of extracted data before a credit report is finalized.

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9. Risks & Questions Throughout Development

- **LLM Stability:** What if Gemini feels exhausted or becomes unresponsive after many uses
- **Data Privacy & Security**: How do we ensure that sensitive financial data (WhatsApp chats, voice notes, sales records) is not leaked or misused?
- **LLM Interpretation Accuracy**: What happens if Gemini misinterprets a WhatsApp chat or voice note?
- **Rural Internet Connectivity**: What happens when a user has poor or no internet connection?

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